

EGERTON



UNIVERSITY

FACULTY OF SCIENCE

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PROJECT REQUIREMENT SPECIFICATION

DOCUMENT(PRS)

FOR

[Topic]

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[program]

COMP 493

[Submission date]

Table of Contents

PROJECT REQUIREMENT SPECIFICATION	1
Table of Figures	3
1.0 Introduction	1
1.1 Project Requirement Specification.....	1
1.2 Background Information	1
1.3 Objectives.....	1
1.4 SCOPE.....	2
1.4.1 Community Reporting:.....	2
1.4.2 Flood Alert Dissemination:.....	2
1.4.3 Educational Features:.....	2
1.4.4 Target Area:	2
1.4.5 System Users:	2
1.5 Out-of-Scope	3
2.0 Requirements.....	3
2.1 Functional Requirements	3
2.2 Non-Functional Requirements	4
3.0 Architecture and Design.....	4
3.1 Timelines	5
3.2 Project Team	7
3.3 Budget and Resources.....	7
3.4 Resources	8
4.0 Risk Management	9
4.1 Risks and Mitigation Strategies	9
5.0 Quality Assurance	10
5.1 QA Components.....	10
5.2 Testing Strategies	11
5.3 Implementation Plan.....	12
5.3.1 Implementation Stages	12
5.3.2 Key Implementation Steps	13
5.4 Maintenance and Support	13

5.5 Support Strategy	14
Conclusion.....	15
Glossary.....	16

Table of Figures

Table 1 Project timelines	6
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1.0 Introduction

1.1 Project Requirement Specification

This is a document that defines the functional and non-functional requirements, objectives, scope, and methodologies for implementing a project. It serves as a map for all stakeholders, defining the technical and operational needs of the system to ensure successful delivery.

The Project Requirement Specification (PRS) document outlines the functional, non-functional, and technical specifications for the Flood Monitoring and Alert System (FMAS). This system is designed to mitigate the impact of flooding in Kenya by improving preparedness, real-time communication, and decision-making.

1.2 Background Information

Kenya experiences annual flooding, particularly in areas like the lower Tana River, the lower Nzoia River at Budalang'i plains, and the lower Nyando River at Kano Plains. The floods experienced in these areas heightened by climate change, cause widespread destruction to lives, livelihoods, and infrastructure, resulting in billions of Kenyan Shillings in losses annually. For instance, the annual flooding in Budalangi often leads to displaced populations, destroyed farmlands, and increased humanitarian needs. Hilly areas traditionally prone to mudslides particularly in Makueni, Nyeri, Muranga and West Pokot Counties are also likely to flood and cause fatalities and cut off road networks.

1.3 Objectives

- Develop a system to forecast flood risks and alert communities in real time.
- Deliver alerts to affected community members via SMS and email.
- Enable affected populations to report incidents anonymously and receive location-based alerts and verification of flooding incidents, fostering inclusivity and participation.
- Educate users on safety protocols via an informative platform eg the websites resource section, email notifications, alert messages.
- Collect detailed flood incident reports, including local area names and descriptions

- Provide step-by-step guidance on flood preparedness and reporting through videos and guides to users.
- Provide step-by-step guidance on flood preparedness and reporting through videos and guides to users.

1.4 SCOPE

1.4.1 Community Reporting:

- Users can submit flood reports with specific local area names and descriptions.
- Allow reports to include multimedia such as images.
- Geotagging to make it easier to identify the actual location a particular flood report submitted by a user.

1.4.2 Flood Alert Dissemination:

- Alerts will be crafted and sent by the admin.
- Focus on delivering alerts to affected users via SMS and email.

1.4.3 Educational Features:

- Provide users with flood preparedness tips and steps to take during emergencies.
- A resource page that provides articles, tools, pdfs and videos to educate the users on how to report a flood incident and the steps to take during a flood incident.

1.4.4 Target Area:

The Budalangi community, which is highly susceptible to flooding.

1.4.5 System Users:

- Community members (residents of Budalangi).
- NDMA as the admin body responsible for verifying reports and issuing alerts.
- Kenya Red Cross as responders for evacuation and humanitarian needs.

1.5 Out-of-Scope

- USSD feature will be considered in future system upgrades.
- The actual disaster response such as evacuation and physical allocation of resources
- The system will exclusively focus on flood-related incidents but there is room to scale to other disaster incidents.
- Deployment of physical resources (e.g., boats, rescue teams) will not be directly managed.

Future Work:

- Expansion to other disaster types (e.g., earthquakes, landslides, fires).
- Integration with emerging technologies like IoT sensors for real-time water level monitoring.

2.0 Requirements

The Flood Monitoring and Alert System (FMAS) requirements are categorized into functional and non-functional specifications to ensure a comprehensive understanding of the system's capabilities and performance expectations.

2.1 Functional Requirements

These requirements describe what the system should do:

- Send alerts via SMS and email to registered users.
- Allow users to set preferences for alert types (severity levels, regions).
- Provide forms for users to report flood incidents.
- Accept image uploads of the flood incidents.
- Include local area names and detailed descriptions in reports.
- Provide a safety map to help user find evacuation routes more easily
- Provide flood preparedness guides and step-by-step safety instructions.

- Host videos and downloadable resources for flood preparedness and reporting procedures.

2.2 Non-Functional Requirements

These requirements describe how the system should perform:

- Support up to 5000 concurrent users during peak times, such as active flood warnings.
- Ensure incident reports are logged within 3 seconds of submission.
- Deliver alerts within 5 seconds of generation.
- Maintain 99.9% uptime, especially during flood seasons.
- Ensure compliance with data privacy regulations, including GDPR and Kenya Data Protection Act.
- Design a user-friendly interface for mobile and web applications.
- Implement modular coding practices for easy updates and feature integration.
- Ensure the system functions across major browsers and devices (desktop and mobile).

3.0 Architecture and Design

The FMAS will employ a robust, modular architecture to ensure reliability, scalability, and ease of maintenance.

Front-End:

- React.js with TypeScript for a responsive web interface.
- Interactive maps, dashboards, charts and user-friendly forms for reporting.

Back-End:

- Node.js with Express.js for handling API requests.
- RESTful APIs for communication between the client and server.
- Axios to send and retrieve data from the API endpoints.

- CORS for communication between the backend and front-end

Database:

- MySQL for structured data storage.
- User profiles, incident reports, alert history.

Hosting:

- Hosting sites (e.g., Netlify for the front-end, and Railway for the backend and database)

Documentation

- Swagger for documenting the backend code.

Visualization:

- **Tools:** Chart.js, D3.js, Open Weather Map or Google Maps API for analytics and geospatial data visualization.

3.1 Timelines

The following table outlines the key milestones and their respective deadlines for the Flood Monitoring and Alert System (FMAS):

Item	Task	Deadline	Details
Project Planning	Requirement gathering and feasibility study, consult the supervisor on requirements elicitation, design PRS, PDD and project proposal	Week 1- 7 (November 2 nd -December 21 st 2024)	Collect Historical data from NDMU and KMD for requirement elicitation. Conduct research on flood and disaster management process

Item	Task	Deadline	Details
Design Completion	UI/UX design for web interfaces, design wireframes for the UI	Week 9- 14 (Jan 2 nd – Jan 31 st 2025)	Design, UI mock for the front-end and landing page
Back-End Development	API and database implementation, swagger documentation	Week 15- 17 Feb 1 st – 15 th	Design the database model, Develop RESTful APIs, routes and controllers.
Front-End Development	Dashboard, reporting, and alerting interfaces	Week 18 – 19 Feb 16 th – 23 rd	Implement responsive designs for cross-device compatibility including a landing page.
Integration and Testing	Connect with external data sources, design test cases and conduct maintenance schedules.	Week 19 -20 Feb 24 th – March 1 st	Test integration with KMD and NDMU.
Develop First Prototype	Present the first prototype to the supervisor	Week 20 - 21 March 1 st – 8 th	Design the first prototype showing all the functionalities of the system
Modify the system based on the feedback	Make the final changes based on the feedback from the supervisor.	Week 21 - 23 March 9 th – March 23 rd	Refine the system based on the user requirements, update the documents (PRS, SDD and proposal)
Full Deployment	Host the system on Netlify and Railway, share the link to users to get their feedback	Week 23 March 25 th to April 1 st	Find hosting sites for the front-end, back-end and the database.
Presentation	Present and defend the project.	Week 26 April 24 th	Present to the panel, defend the project and submit the final working copy of the system and its documentation.

Table 1 Project timelines

3.2 Project Team

The development and implementation of the Flood Monitoring and Alert System (FMAS) will involve the following key participants:

Project Developer:

Role: Responsible for designing, developing, testing, and deploying the FMAS. This includes coding the front-end and back-end, integrating APIs, and ensuring the system meets the outlined requirements.

Stakeholders:

Role: Provide insights, feedback, and guidance to ensure the system aligns with real-world disaster management needs. They also assist in defining requirements and validating the functionality of the system.

End Users:

Role: Represent the communities and institutions that will use the FMAS. They provide feedback on usability, identify gaps, and suggest improvements. End users are critical in testing the system during the pilot phase.

Flood-prone communities like the locals at Budalangi area, NMDU and responders (Kenya Red Cross).

3.3 Budget and Resources

The successful implementation of the Flood Monitoring and Alert System (FMAS) requires careful planning and allocation of resources. Below is the detailed budget and resources breakdown for the project:

Item	Description	Cost Estimate (KES)
Hardware Resources	Devices for development and testing (e.g., laptops, servers)	80,000

Item	Description	Cost Estimate (KES)
Software Licenses	Tools for development, database management, and APIs	20,000
Cloud Hosting	Infrastructure for hosting the FMAS platform	5,000/month
Communication Tools	SMS Gateway integration for alert notifications	5000
Research and Data	Data collection from NDMU, KMD and Kenya Red Cross	1000
Development Costs	Coding, testing, and deployment of the system	N/A
Awareness Campaigns	Publicizing FMAS to target users eg poster, seminars, local meetups and radio broadcasting.	5,000
Miscellaneous Costs	Unforeseen expenses	5,000
Total Estimated Cost		121,000 KES

3.4 Resources

Hardware Resources:

- Laptop for development and testing.
- Server for hosting the system post-deployment.

Software Tools:

- IDEs (e.g., Visual Studio Code).
- Database management tools (e.g., MySQL).
- APIs for geospatial and flood-risk data integration.

Third-party Services:

- SMS gateways and email services for alert dissemination.

4.0 Risk Management

A critical part of the FMAS is identifying, mitigating, and monitoring risks to ensure successful implementation and long-term sustainability.

4.1 Risks and Mitigation Strategies

Data Reliability and Accuracy:

- **Risk:** Receiving inaccurate or incomplete flood-related data may impact the system's effectiveness.
- **Mitigation:** Partnering with credible institutions like Kenya Red Cross, Kenya Meteorological Department (KMD), and NMDU for verified data.

Connectivity Challenges:

- **Risk:** Poor internet connectivity in remote flood-prone areas could hinder system functionality.
- **Mitigation:** Incorporate offline capabilities like SMS-based reporting and alerts.

User Adoption:

- **Risk:** Target users might be unaware of the system or reluctant to adopt new technologies.
- **Mitigation:** Conduct awareness campaigns and training sessions to familiarize users with FMAS features. Hold seminars and collaborate with local authorities such as chiefs to make people aware of the platform. Use posters and radio broadcasts to make users aware of the system.

Budget Constraints:

- **Risk:** Limited funding may delay implementation.
- **Mitigation:** Prioritize core functionalities during the initial rollout and seek additional funding or partnerships if required.

System Security:

- **Risk:** Unauthorized access to the system could compromise sensitive data.
- **Mitigation:** Implement robust security protocols, such as encryption and user authentication mechanisms.

Maintenance Challenges:

- **Risk:** Lack of regular maintenance may lead to system downtime or inefficiencies.
- **Mitigation:** Develop a clear maintenance plan and schedule periodic system audits.

Scalability Issues:

- **Risk:** Future expansion to cover additional regions or features might strain resources.
- **Mitigation:** Use scalable cloud hosting solutions and modular development practices.

5.0 Quality Assurance

Ensuring the Flood Monitoring and Alert System (FMAS) meets the highest standards of functionality, reliability, and user satisfaction is paramount.

5.1 QA Components

System Reliability:

- FMAS must deliver accurate alerts and reliable data under all conditions.
- 99% uptime with minimal false positives in flood alerts.

Response Time:

- Timely notifications are critical during flood emergencies.
- Alerts dispatched within 30 seconds of data collection and validation.

User Interface (UI) Usability:

- The interface should be intuitive for both technical and non-technical users.

- Achieve an average usability score of 4.5/5 in user feedback surveys.

Data Security:

- Protect sensitive user and organizational data.
- Compliance with ISO 27001 and Kenya Data Protection Act (2019).

5.2 Testing Strategies

Unit Testing:

- Validate individual modules, such as data collection, alert dissemination, and registration.

Integration Testing:

- Test interactions between FMAS components, including APIs and databases.

Performance Testing:

- Assess the system under high load conditions, simulating floods in multiple regions.

User Acceptance Testing (UAT):

- Conduct hands-on testing with Kenya Red Cross personnel and community representatives.

Security Testing:

- Verify encryption, authentication, and vulnerability protections against cyberattacks.

5.3 Implementation Plan

The successful rollout of FMAS requires a structured approach, involving development, testing, deployment, and user onboarding. Below is the detailed implementation plan:

5.3.1 Implementation Stages

Development Stage:

- Design and coding of the system's core functionalities: flood alerts, reporting, geospatial mapping, and analytics.
- Duration: 3 months.

Testing Stage:

- Conduct unit, integration, and user acceptance testing.
- Involve stakeholders, such as the NDMU and Kenya Red Cross, for feedback and adjustments.
- Duration: 1 month.

Pilot Deployment:

- Roll out FMAS in a flood-prone region (e.g., Budalang'i).
- Collect real-world feedback from users to refine the system.
- Duration: 1 month.

Full Deployment:

- Launch FMAS in all target regions across Kenya.
- Train local authorities and community members on system use.
- Duration: 3 weeks.

Monitoring and Support:

- Establish a dedicated support team to address issues and provide updates.
- Duration: It will be Ongoing task.

5.3.2 Key Implementation Steps

Stakeholder Engagement:

- Collaborate with NDMU, Kenya Red Cross, KMD, and community representatives for data and insights.

User Training:

- Organize workshops and provide materials (e.g., guides, videos) for end-users.

Awareness Campaign:

- Launch campaigns via social media, SMS, and community meetings to inform users about FMAS.

Maintenance Setup:

- Schedule routine maintenance and updates to ensure the system's long-term reliability.

5.4 Maintenance and Support

Ensuring the Flood Monitoring and Alert System (FMAS) remains effective and reliable over time is critical. A comprehensive maintenance and support strategy will be implemented to address potential issues, improve system performance, and enhance user satisfaction.

Regular System Updates:

- Deploy updates to address bugs, integrate new features, and enhance system performance.
- Frequency: as needed.

Server and Database Maintenance:

- Perform routine checks on servers and databases to ensure data integrity and prevent downtime.

- Frequency: weekly.

Security Enhancements:

- Conduct regular security audits and implement patches to safeguard user data.
- Frequency: after major security updates.

User Feedback Integration:

- Collect and analyze feedback from users (e.g., Kenya Red Cross, community members, NDMU) to prioritize improvements.
- Frequency: Continuous.

Data Backup and Recovery:

- Implement automated data backup processes and test recovery mechanisms.
- Frequency: Weekly.

5.5 Support Strategy

Training and Onboarding:

- Provide periodic training sessions for new users, including local authorities and community representatives.

Documentation and Guides:

- Maintain comprehensive user guides, FAQs, video tutorials and troubleshooting documentation accessible via the FMAS platform.

Community Engagement:

- Host forums and workshops to educate users about system capabilities and gather suggestions for improvement.

Conclusion

The Flood Monitoring and Alert System (FMAS) represents a transformative step in disaster preparedness and response in Kenya. By integrating real-time data analytics, geospatial mapping, and community-driven reporting, FMAS addresses critical gaps in current flood management practices. The system's scalability, offline capabilities, and user-friendly design ensure that it meets the diverse needs of stakeholders, including the Kenya Red Cross, government agencies, and vulnerable communities.

This PRS document provides a roadmap for the development, deployment, and sustainability of FMAS. From detailed timelines and budgets to risk management and quality assurance measures, every aspect has been meticulously planned to ensure the project's success. With proper implementation, FMAS will significantly enhance flood response efficiency, save lives, and mitigate the devastating impacts of floods.

Glossary

Flood Monitoring and Alert System (FMAS): A digital system designed to collect, analyze, and disseminate real-time flood-related data to stakeholders and communities.

National Disaster Management Unit (NDMU): Disaster management unit which deals with disaster risk management, preparedness and response. Their mission is to effectively prepare and respond to disasters, manage recovery and mitigation efforts in Kenya

Kenya Red Cross (KRC): A humanitarian organization providing disaster relief, health services, and community support in Kenya.

Geospatial Mapping: The process of using geographic data to create maps that assist in monitoring and managing flood-prone areas.

Real-Time Data: Information that is collected and processed instantly, enabling quick decision-making during emergencies.

API (Application Programming Interface): A set of programming instructions allowing different software systems to communicate with each other.

Quality Assurance (QA): Procedures to ensure the system meets specified performance and reliability standards.

Stakeholders: Individuals or organizations with an interest in the FMAS, including developers, users, and funding bodies.

CORS: Cross-origin resource sharing (CORS) is a mechanism for integrating applications. CORS defines a way for client web applications that are loaded in one domain to interact with resources in a different domain.